Turtle Games Customer Analytics: Technical Report

1. Background and objectives

Turtle Games aims to improve sales performance by understanding customer behaviour, loyalty and feedback. Using 2,000 customer records, this report addresses four questions: how loyalty points are accumulated, how customers can be segmented, what reviews reveal, and whether the loyalty-points data suit predictive modelling. Turtle Games aims to improve sales performance by understanding customer behaviour, loyalty and feedback, with the analysis supporting more targeted marketing and customer-retention decisions.

2. Analytical approach

2.1 Data preparation

The dataset holds 2,000 records and 11 variables, with no missing values or duplicates. Language and platform each had a single value and were dropped, so conclusions apply only to that language and platform. Columns were renamed, and the product ID was excluded from modelling because it is a label, not a quantity. Free-text reviews were analysed separately with NLP.

2.2 Tools

R handled cleaning, exploratory analysis and regression diagnostics, while Python handled model comparison, clustering and NLP. The linear baseline was fitted in both languages as a cross-check.

2.3 Modelling and validation

Each supervised model was trained on 70% of records and scored on the remaining 30%, so results reflect unseen customers. A linear regression on age, remuneration and spending score was the interpretable baseline (James et al., 2021), with variance inflation factors of 1.00 to 1.06. The model was extended with a remuneration × spending score interaction after the residual plot showed curvature. Decision trees were chosen because they capture non-linearity and interactions without distributional assumptions. An unpruned tree overfitted (training R^2 1.000), so depth and minimum leaf size were tuned by five-fold cross-validation on training data (Kohavi, 1995), keeping the simplest tree within 0.01 of the best cross-validated R^2 (depth 8, 108 leaves). A 300-tree Random Forest tested whether an ensemble added accuracy (Breiman, 2001).

2.4 Segmentation and text

K-means grouped customers on remuneration and spending score, which share similar ranges and so were left unscaled; elbow and silhouette results for k = 1 to 10 gave k = 5. Reviews were lower-cased, stripped of punctuation and digits, filtered with NLTK stop words and scored with TextBlob, with polarity above 0.1 or below -0.1 defining positive and negative.

2.5 Limitations.

Regression diagnostics indicated non-linearity and heteroscedasticity. Results also depend on one train/test split, K-means uses only two variables, and TextBlob cannot reliably capture context or sarcasm.

3. Visualisation and insights

3.1 Customer behaviour and loyalty

Loyalty points ranged from 25 to 6,847, with a mean of 1,578 above the median of 1,276. Skewness of 1.46 and excess kurtosis of 1.71 indicate strong right-skewness and heavier tails than normal (Shapiro-Wilk, $p < 0.001$; Figure 1). The IQR rule flagged 266 loyalty-point observations (13.3%) as potential outliers; these were retained as plausible high-value customers. Regression assumes normal residuals, not a normal target, and trees assume no distribution, so the data suit prediction but favour non-linear models.

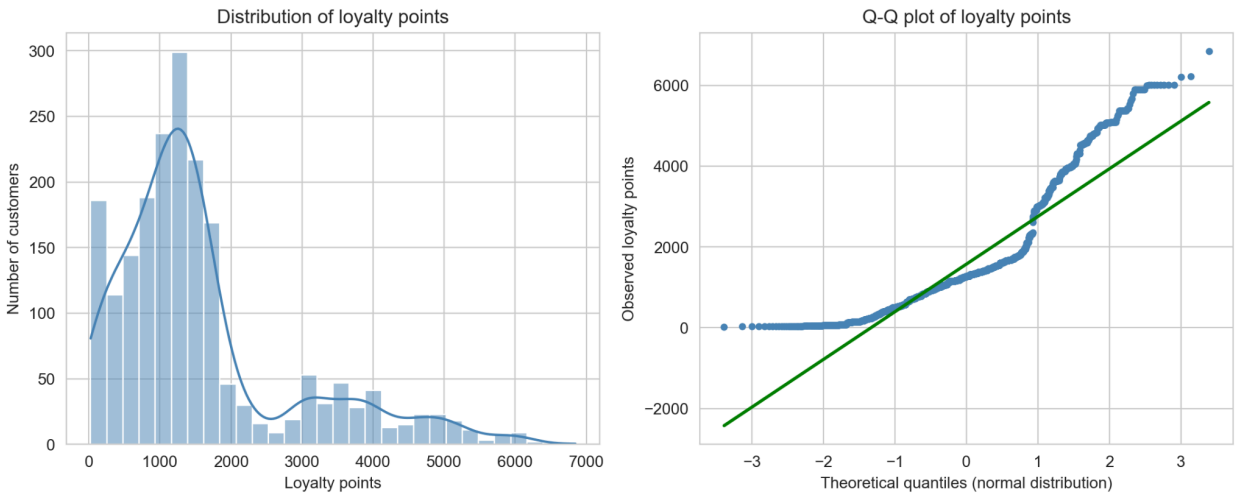


Figure 1. Distribution of customer loyalty points, showing the strong positive skew and the concentration of customers at lower loyalty-point totals, with a smaller number of customers accumulating substantially higher points.

Loyalty points correlated moderately with remuneration (r = 0.62) and spending score (r = 0.67) but not with age (r = -0.04); remuneration and spending score were almost uncorrelated (r = 0.01). Figure 2 shows points spreading out at high spending scores, so a straight line understates the pattern.

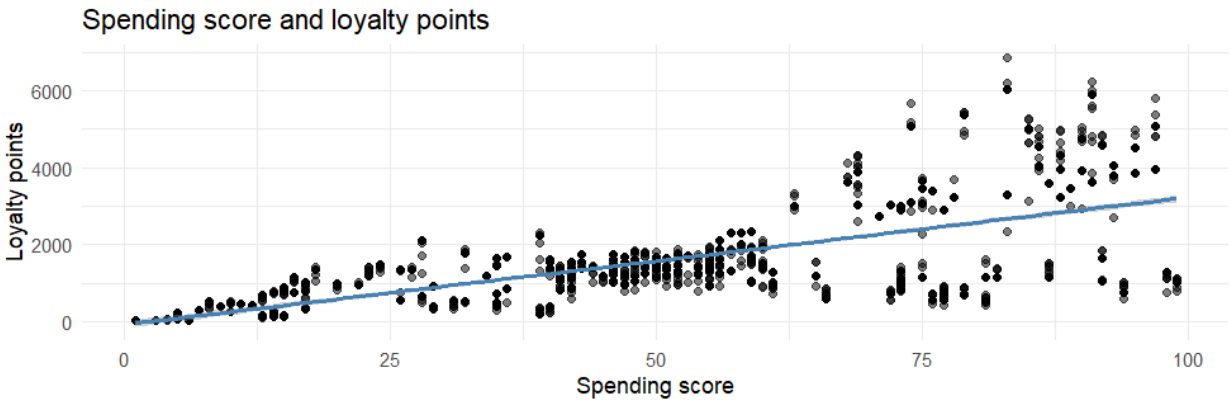


Figure 2. Relationship between customer spending score and loyalty points, showing the positive association between higher spending scores and greater loyalty-point accumulation.

The linear baseline gave test R^2 values of 0.841 in R and 0.830 in Python, providing a useful cross-check. Table 1 compares Python models on one split: an interaction term raised test R^2 to 0.983, the pruned tree reached 0.984 and the Random Forest 0.996.

Table 1: Model performance on the held-out 30% of customers (RMSE and MAE in loyalty points)

Model	Train R^2	Test R^2	Test RMSE	Test MAE
Random forest	0.999	0.996	76	26
Decision tree, unpruned	1.000	0.995	90	33
Decision tree, pruned (depth 8)	0.991	0.984	159	94
Linear regression + interaction	0.982	0.983	166	119
Linear regression	0.844	0.830	525	398

Diagnostics confirmed the linear model's limits: the residual plot curved (Figure 3), the Breusch-Pagan test indicated heteroscedasticity ($p < 0.001$) and residuals were not normal (Shapiro-Wilk $p < 0.001$). It is therefore used as an interpretable baseline only.

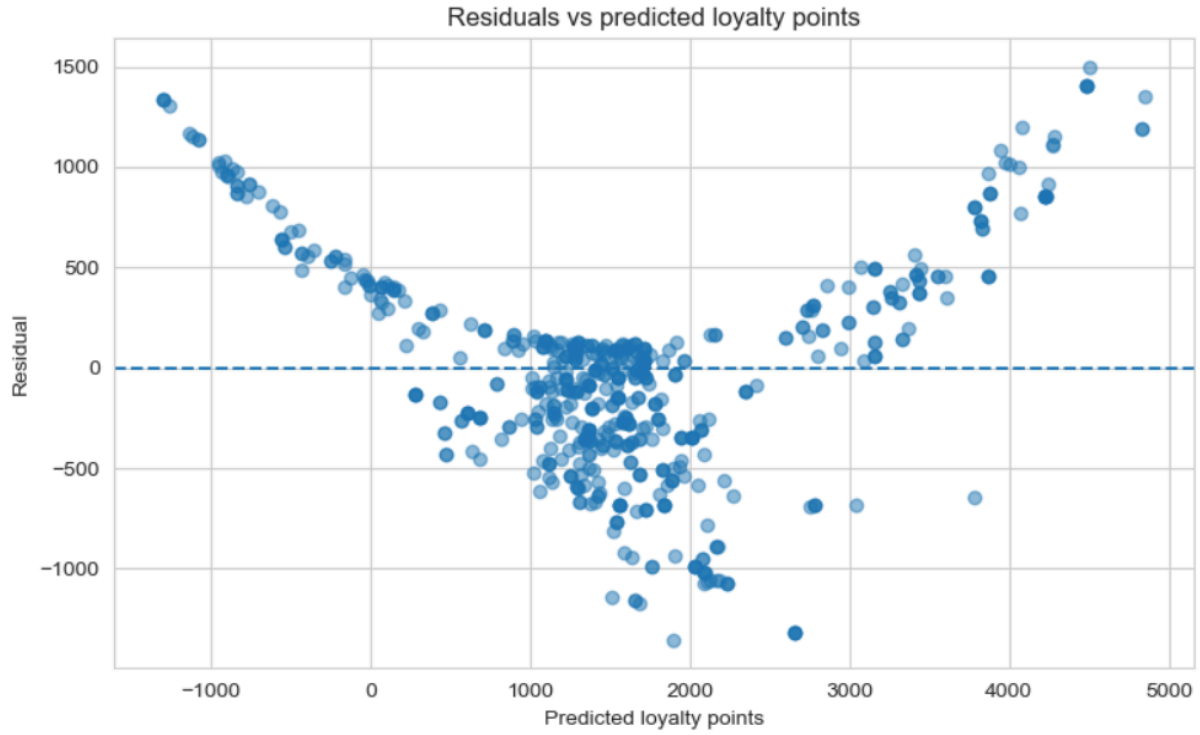


Figure 3. Residuals versus predicted loyalty points for the linear regression model, showing departures from a completely random pattern that indicate remaining non-linearity and variation in residual spread.

Remuneration (0.50) and spending score (0.49) accounted for almost all of the tree's feature importance; age contributed 0.01, gender and education almost nothing. A test R^2 of 0.996 suggests loyalty points may be calculated from these two variables, so the models may be reconstructing the scheme's rules rather than uncovering behaviour.

3.2 Customer segmentation

K-means produced five segments (Figure 4; silhouette = 0.583), indicating reasonable cluster separation (Rousseeuw, 1987). The largest, Mid income / Mid spend, holds 38.7% of customers; the others hold 13% to 18% each. High income / Low spend customers (16.5%) appear able to spend more but do not, making them the clearest target for personalised offers; High income / High spend customers (17.8%) suit retention rewards. The scatterplot shows the groups directly, but two variables cannot capture age or product preferences, and segments depend on the chosen k.

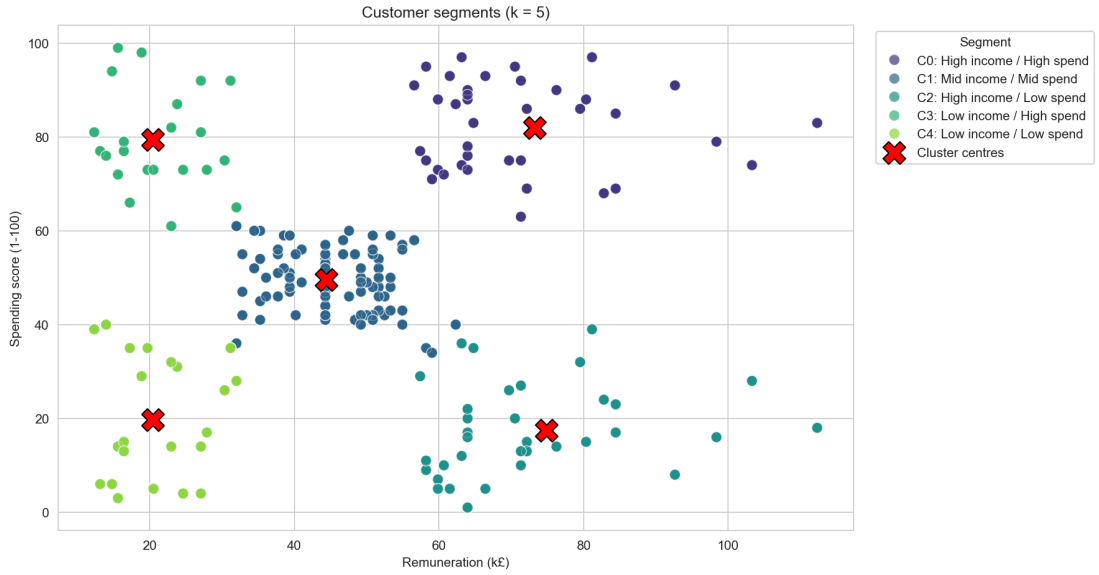


Figure 4. Customer segmentation based on remuneration and spending score using K-means clustering (k = 5), identifying distinct high-, medium- and low-income spending groups for potential targeted marketing.

3.3 Customer reviews and sentiment

Review text was analysed to see what customers value and where products disappoint. Figure 5 shows the overall sentiment distribution, with most reviews classified as positive, but TextBlob scores 'game' at -0.4, illustrating how individual word scores can mislead and highlighting the limitations of automated sentiment classification (Pang et al., 2002). Sentiment was almost unrelated to spending score, remuneration, age or loyalty points (all $|r| < 0.05$). Negative reviews highlighted complicated instructions, incomplete products, boredom and poor value.

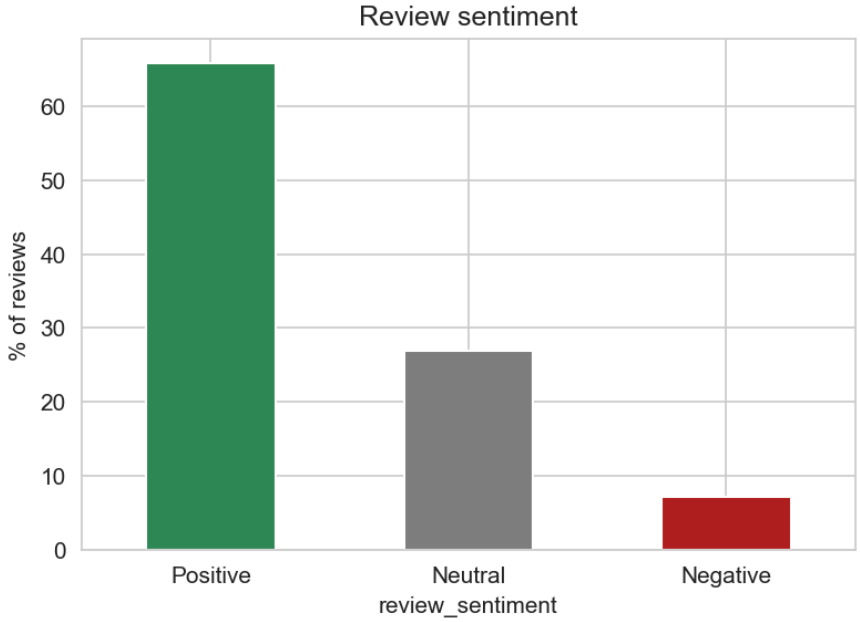


Figure 5. Distribution of customer review sentiment, showing that 65.8% of reviews were classified as positive, compared with 27.0% neutral and 7.2% negative reviews.

4. Patterns, predictions and recommendations

Loyalty points are driven almost entirely by remuneration and spending score (importance 0.50 and 0.49; age 0.01), acting together: an interaction term lifted linear test R^2 from 0.83 to 0.98. The tree and Random Forest predict unseen customers' points with typical errors of 76 to 159 points (5 to 10% of the 1,578 average), accurate enough to rank customers for campaigns. Recommended actions, in priority order:

Trial personalised offers on High income / Low spend customers (16.5%) against a randomised control group, comparing loyalty points per customer after a fixed period.

Investigate the instruction and kit-completeness complaints behind the 7.2% negative reviews.

Confirm how points are calculated, then retrain the model on new data and track its error.

References:

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